Practical no. 19

Name :- Danish Altaf Solkar **Enrolment no. :-24111590805**

Class :- CO3K – B **Subject :- DMS**

Aim:- Implement PL/SQL program using Iterative Statements

Program :-

Run SQL Command Line

```
SQL> Declare
  2  counter NUMBER :=1;
  3  Begin
  4  dbms_output.put_line('PL/SQL basic loop execution');
  5  Loop
  6  dbms_output.put_line('This is iteration number '|| counter);
  7  If counter = 4 THEN
  8  Exit;
  9  End if;
 10  counter := counter + 1;
 11  End loop;
 12  End;
 13  /
PL/SQL basic loop execution
This is iteration number 1
This is iteration number 2
This is iteration number 3
This is iteration number 4

PL/SQL procedure successfully completed.
```

Program :-

Run SQL Command Line

```
SQL> Declare
  2  counter number;
  3  Begin
  4  dbms_output.put_line('PL/SQL for loop execution');
  5  For counter in 1..5 loop
  6  dbms_output.put_line('Counter value:'||counter);
  7  End loop;
  8  End;
  9  /
PL/SQL for loop execution
Counter value:1
Counter value:2
Counter value:3
Counter value:4
Counter value:5

PL/SQL procedure successfully completed.
```

Program :-

Run SQL Command Line

```
SQL> Declare
  2  counter number;
  3  Begin
  4  dbms_output.put_line('PL/SQL reverse for loop execution');
  5  For counter in reverse 1..10 loop
  6  dbms_output.put_line ('Counter value:'||counter);
  7  End loop;
  8  End;
  9  /
PL/SQL reverse for loop execution
Counter value:10
Counter value:9
Counter value:8
Counter value:7
Counter value:6
Counter value:5
Counter value:4
Counter value:3
Counter value:2
Counter value:1

PL/SQL procedure successfully completed.
```

Program :-

```
SQL> Declare
  2  counter number := 1;
  3  Begin
  4  dbms_output.put_line('PL/SQL while loop execution');
  5  While counter <= 5
  6  Loop
  7  dbms_output.put_line ('Counter value:'||counter);
  8  counter := counter+1;
  9  End loop;
 10  End;
 11  /
PL/SQL while loop execution
Counter value:1
Counter value:2
Counter value:3
Counter value:4
Counter value:5

PL/SQL procedure successfully completed.
```

Practical related questions :-

1) Program :-

```
SQL>  Declare
  2  vnum number := &vnum;
  3  vresult number;
  4  Begin
  5  dbms_output.put_line('Multiplication table of '|| vnum);
  6  For n in 1..10
  7  Loop
  8  vresult := vnum * n;
  9  dbms_output.put_line (vnum || '*' ||n|| '=' || vresult);
 10  End loop;
 11  End;
 12  /
Enter value for vnum: 5
old   2: vnum number := &vnum;
new   2: vnum number := 5;
Multiplication table of 5
5*1=5
5*2=10
5*3=15
5*4=20
5*5=25
5*6=30
5*7=35
5*8=40
5*9=45
5*10=50

PL/SQL procedure successfully completed.
```

2) Program :-

```
SQL> Declare
  2  vnum number := &vnum;
  3  vfact number := 1;
  4  vcounter number := 1;
  5  Begin
  6  While vcounter <= vnum
  7  Loop
  8  vcounter := vcounter + 1;
  9  End loop;
 10  dbms_output.put_line ('Factorial of ' ||vnum|| ' is : ' || vcounter);
 11  End;
 12  /
Enter value for vnum: 9
old   2: vnum number := &vnum;
new   2: vnum number := 9;
Factorial of 9 is : 10

PL/SQL procedure successfully completed.
```

3) **Program** :-

Run SQL Command Line

```
SQL> Declare
i number;
j number;
flag boolean;
Begin
dbms_output.put_line('Prime numbers between 1 and 50 :- ');
For i in 2..50
Loop
flag := true;
if i > 2 and mod(i,2) = 0 THEN
flag := false;
Else
For j in 2..trunc(sqrt(i))
loop
If mod(i,j) = 0 THEN
flag := false;
Exit;
End if;
End loop;
End if;
If flag THEN
dbms_output.put_line (i);
End if;
End loop;
End;
/
```

```
Prime numbers between 1 and 50 :-
2
3
5
7
11
13
17
19
23
29
31
37
41
43
47

PL/SQL procedure successfully completed.
```